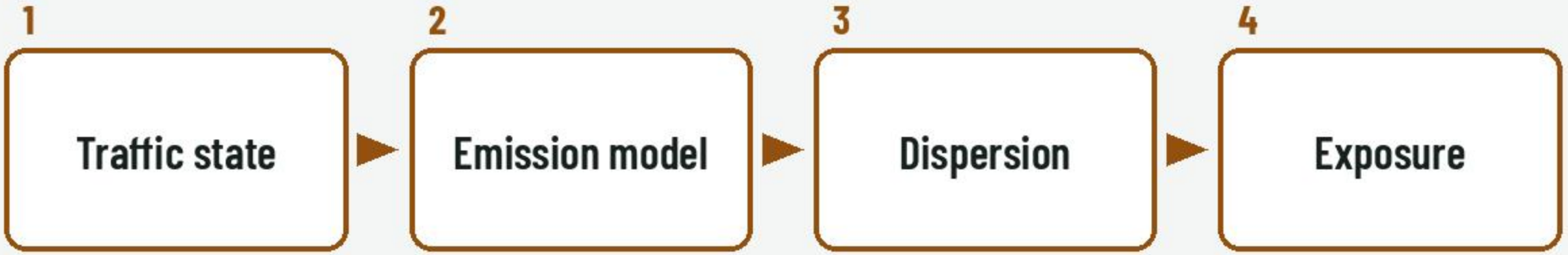
APPLICATIONS

City twins

Module 11 · Environment, smart cities and decision support

Transport Twin Study Path

From traffic to air quality



Emission models

- HBEFA, COPERT (Europe), MOVES (US)
- SUMO includes HBEFA-based emission classes
- Validate with air-quality monitors

City-scale twins

- 3D city models plus live data
- Platforms: FIWARE, Snap4City, cloud twin services
- Examples: Virtual Singapore, Helsinki

Interfaces for decisions

- Summary first, detail on demand
- Show uncertainty, not just a number
- Explain why the twin recommends an action

Remember

A twin nobody uses is a model

Design for the operator, the planner and the public